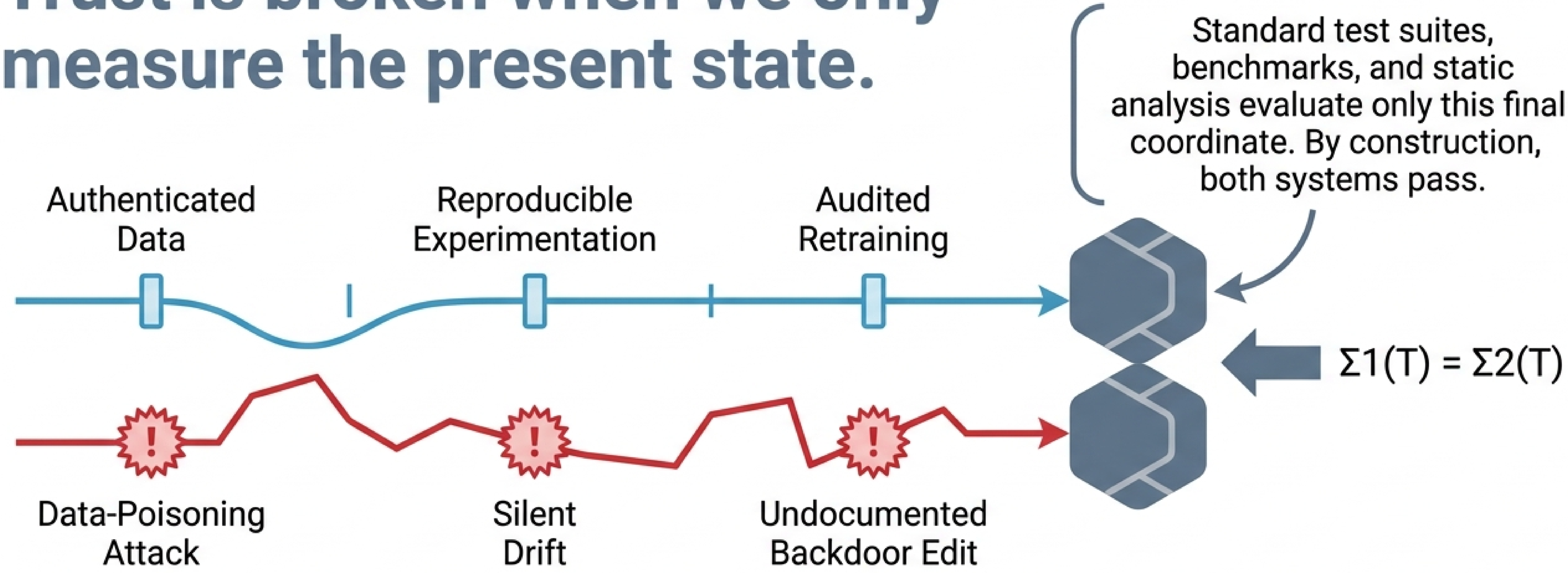
The Architecture of Admissibility

Trust as a Property of System Histories

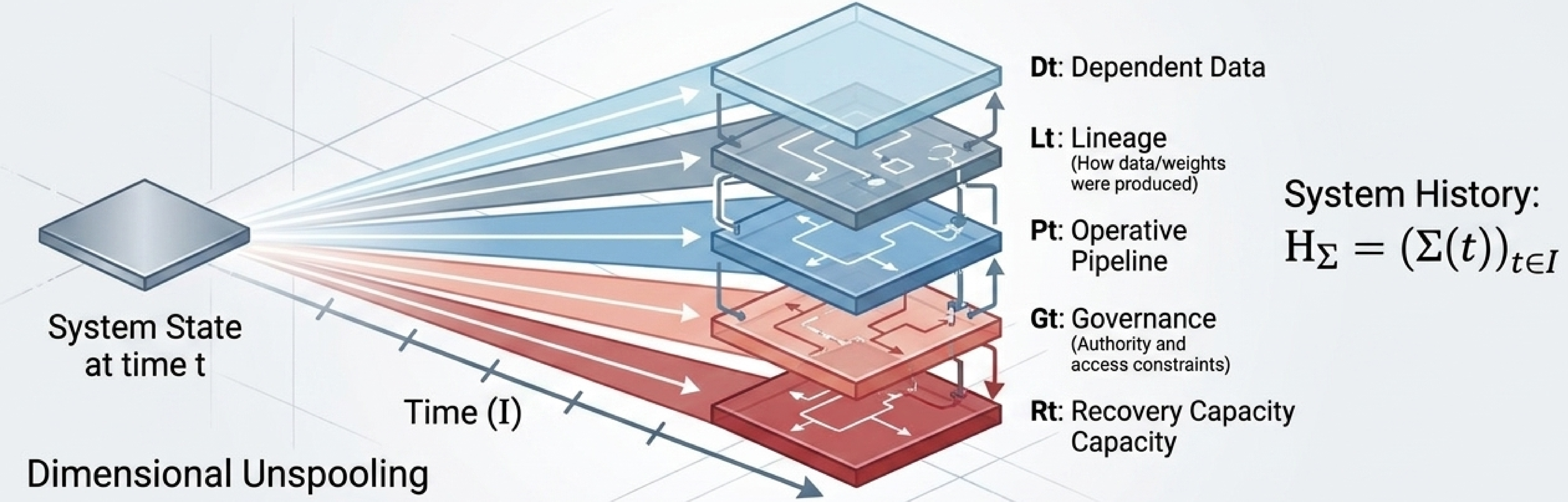
Based on research by Flyxion (July 2026)

Trust is broken when we only measure the present state.



The Identical-Endpoint Problem: State alone underdetermines history. A compromised system can perfectly masquerade as a secure one if the evaluator only inspects the present.

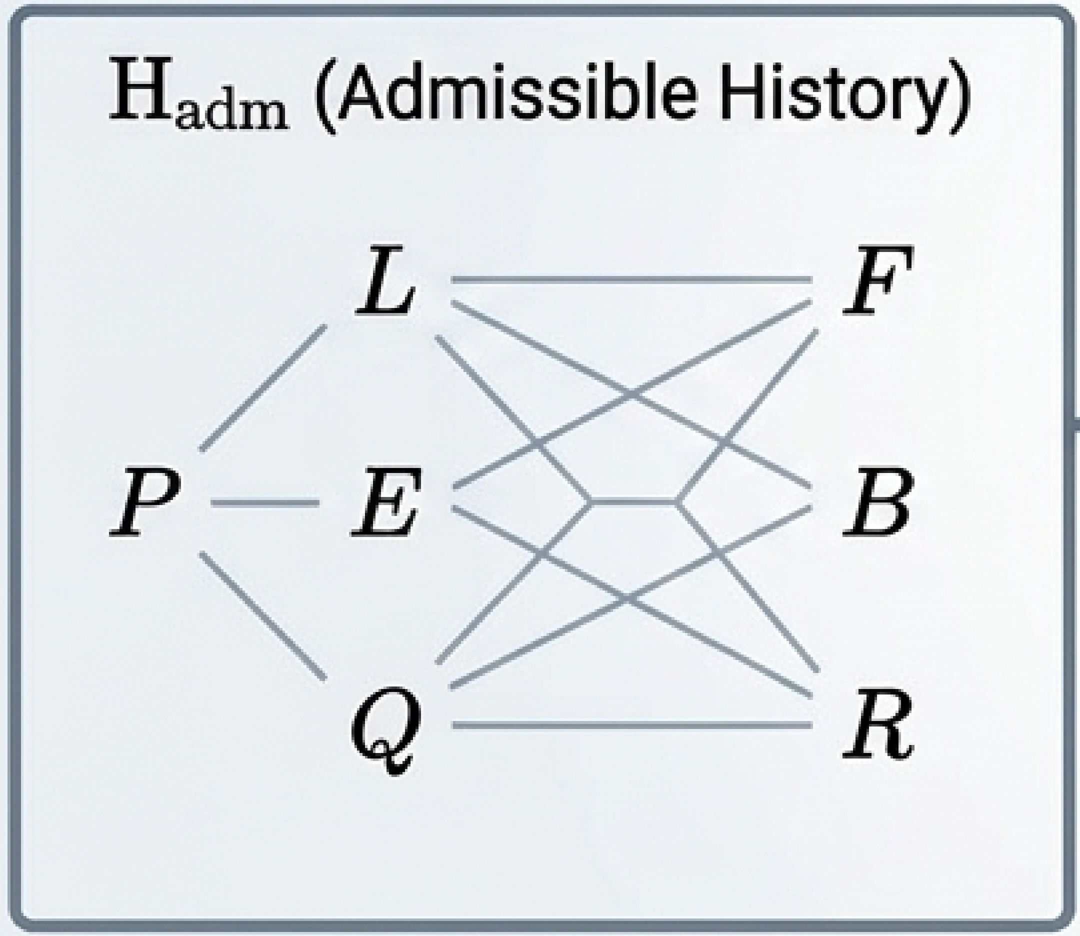
A system reduced to its parameters has already had its history discarded.



History is deliberately not a set of states. It is the ordered, temporal trajectory. Trust requires evaluating admissibility over the entire trajectory, not a single snapshot: $\mathrm{A}(\mathrm{H}_\Sigma) \in \{0, 1\}$.

The industry is already building history-based trust out of practical necessity.

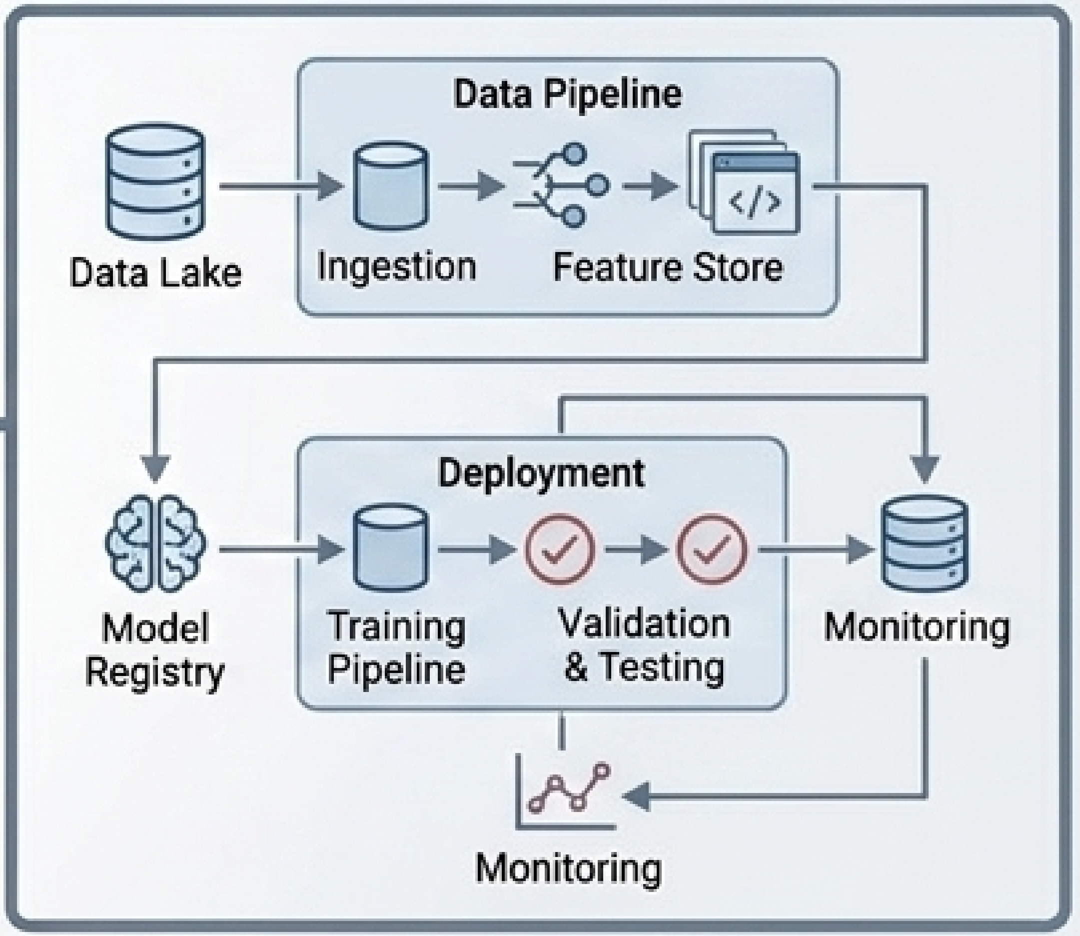
Formal Admissibility Theory (Flyxion)



Formal Admissibiity Theory (Flyxion)

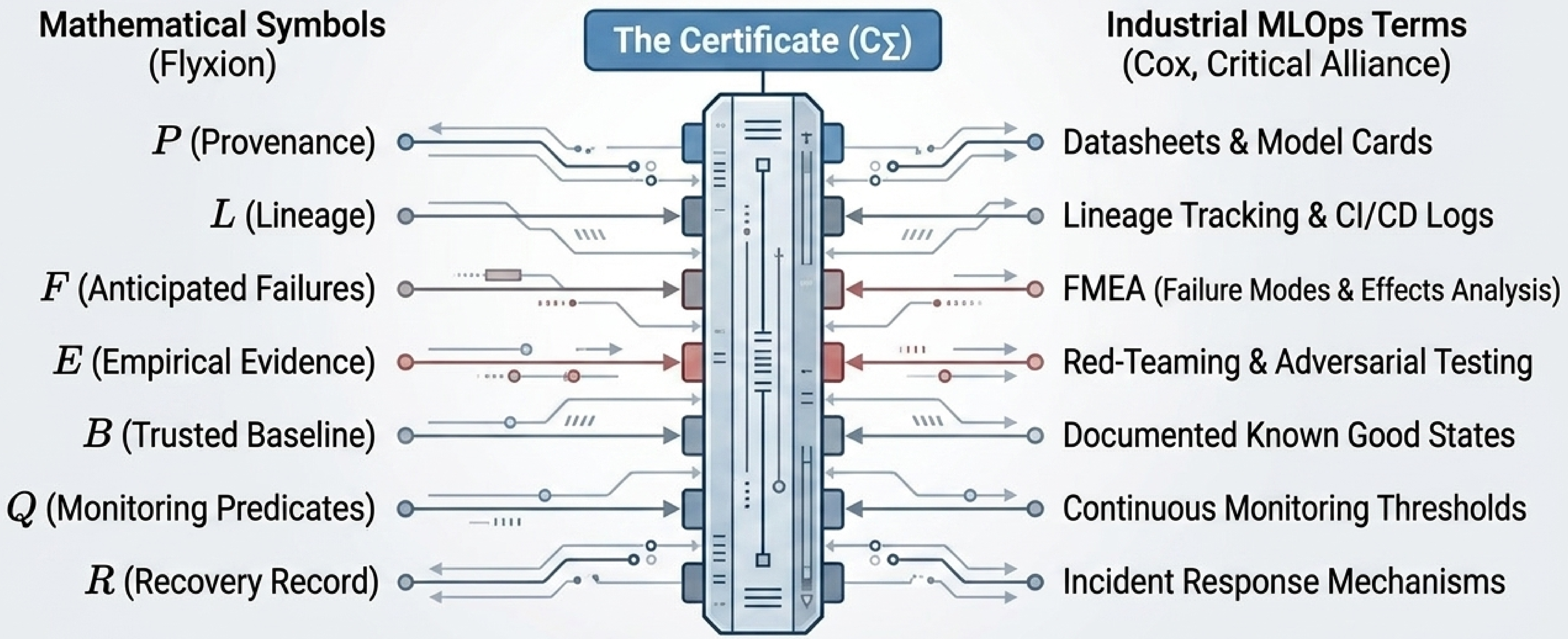
Cox's industrially-motivated MLOps framework independently arrives at nearly every component of formal admissibility without recognizing them as instances of a single mathematical structure.

Securing AIML Systems in the Age of Information Warfare (Susanna Cox, Critical Alliance)



Pragmatic MLOps Architecture (v. Cox., Critical, Alliance)

Bridging formal constraint theory and practical MLOps engineering.

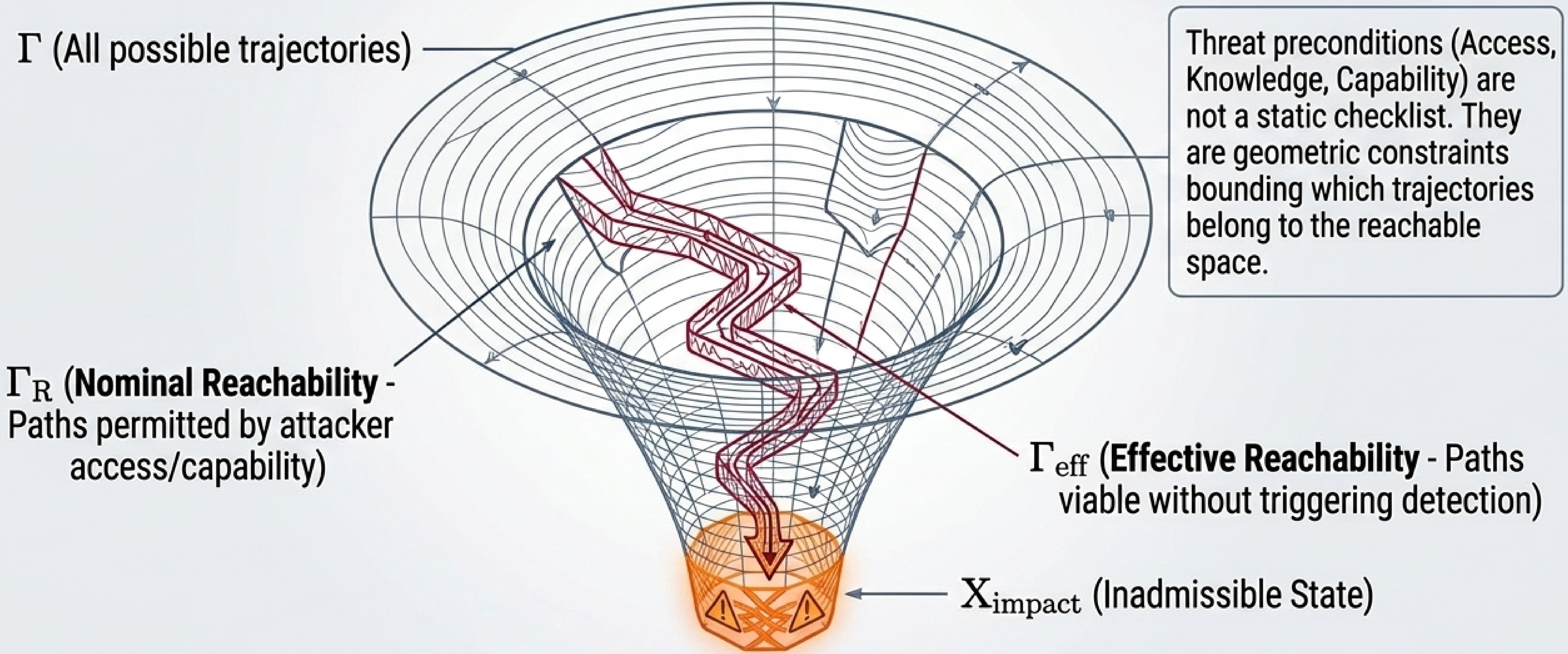


The Hierarchy of Recovery: Why retraining rarely resolves a compromise.

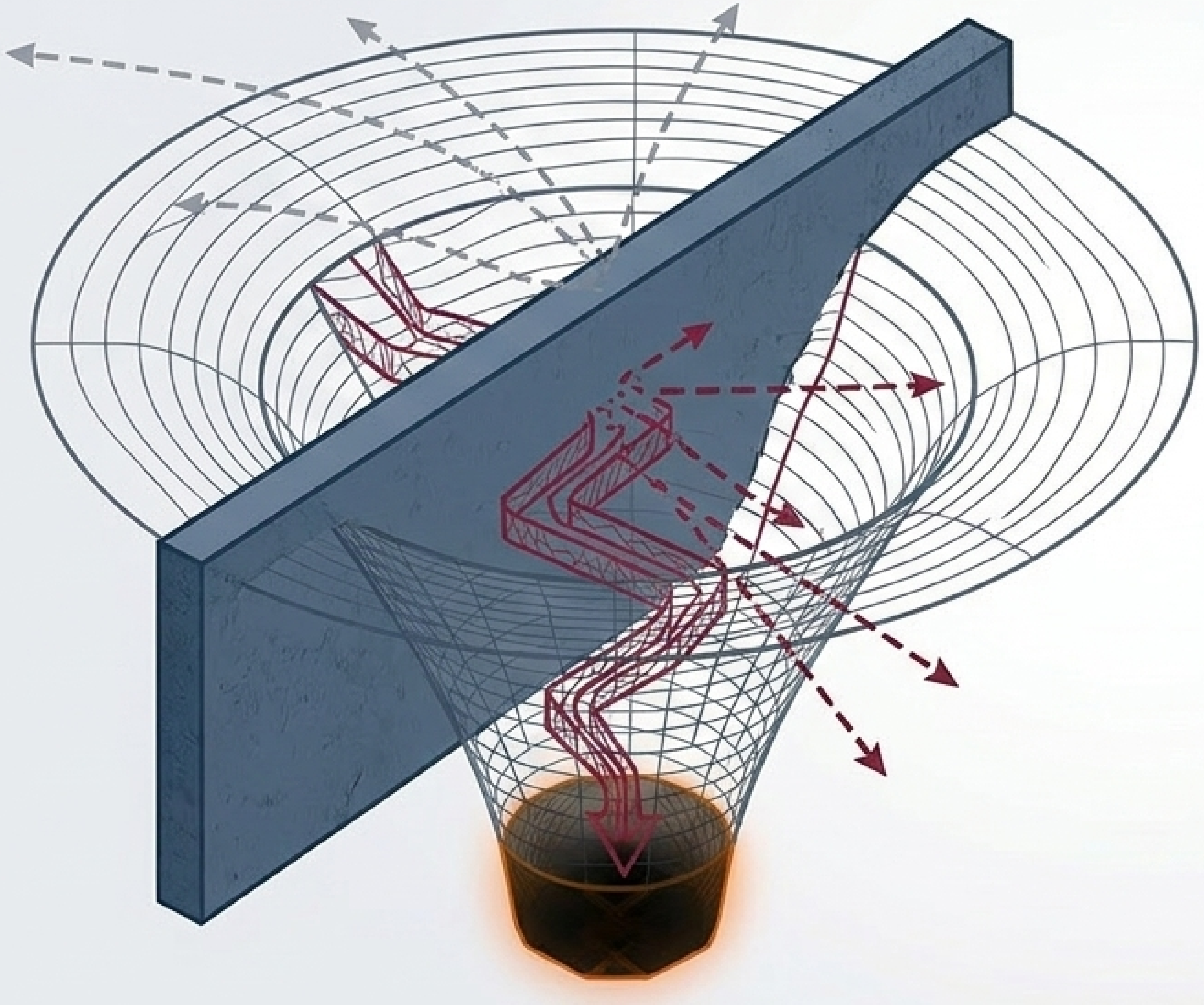
Operation	Action Taken	Target Restored	Vulnerability Status
Rollback	Replaces current parameters (θ_t) with stored prior parameters (θ_t').	Restores an Artifact.	Underlying vulnerability is untouched.
Reconstruction	Retrains the model against previously validated data and pipeline configurations.	Restores a State (Σ_t').	Attacker trap remains active in the environment.
Repair	Re-derives the state subject to a preservation constraint, diagnosing and closing the mechanism of compromise.	Restores Admissibility to the History.	Forward trajectory is secure.

Proposition 1: Rollback $\subset$ Reconstruction $\subset$ Repair. Only repair restores trust.

An attack is the successful traversal of a trajectory into inadmissibility.



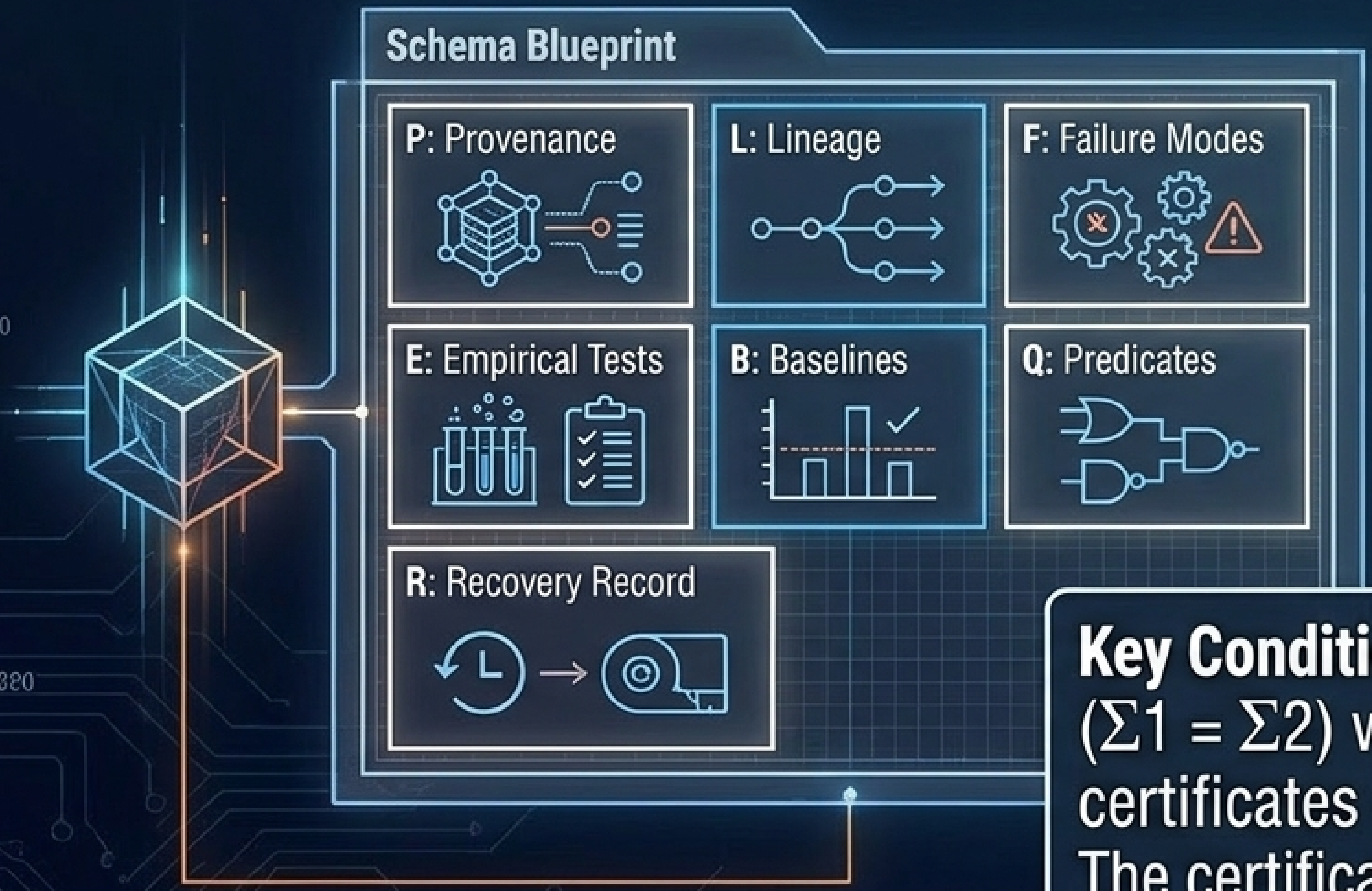
Defense is trajectory elimination, not perimeter hardening.



- **Blue succeeds if and only if $\Gamma_{\mathrm{eff}} \cap \mathrm{X}_{\mathrm{impact}} = \emptyset$.**
- Hardening one access route **while leaving a disjunctive alternative open accomplishes nothing.**

Repair (Defined): Closing the relevant portion of Γ_{eff}, not merely relocating the system back to a starting point for the attacker to re-traverse.

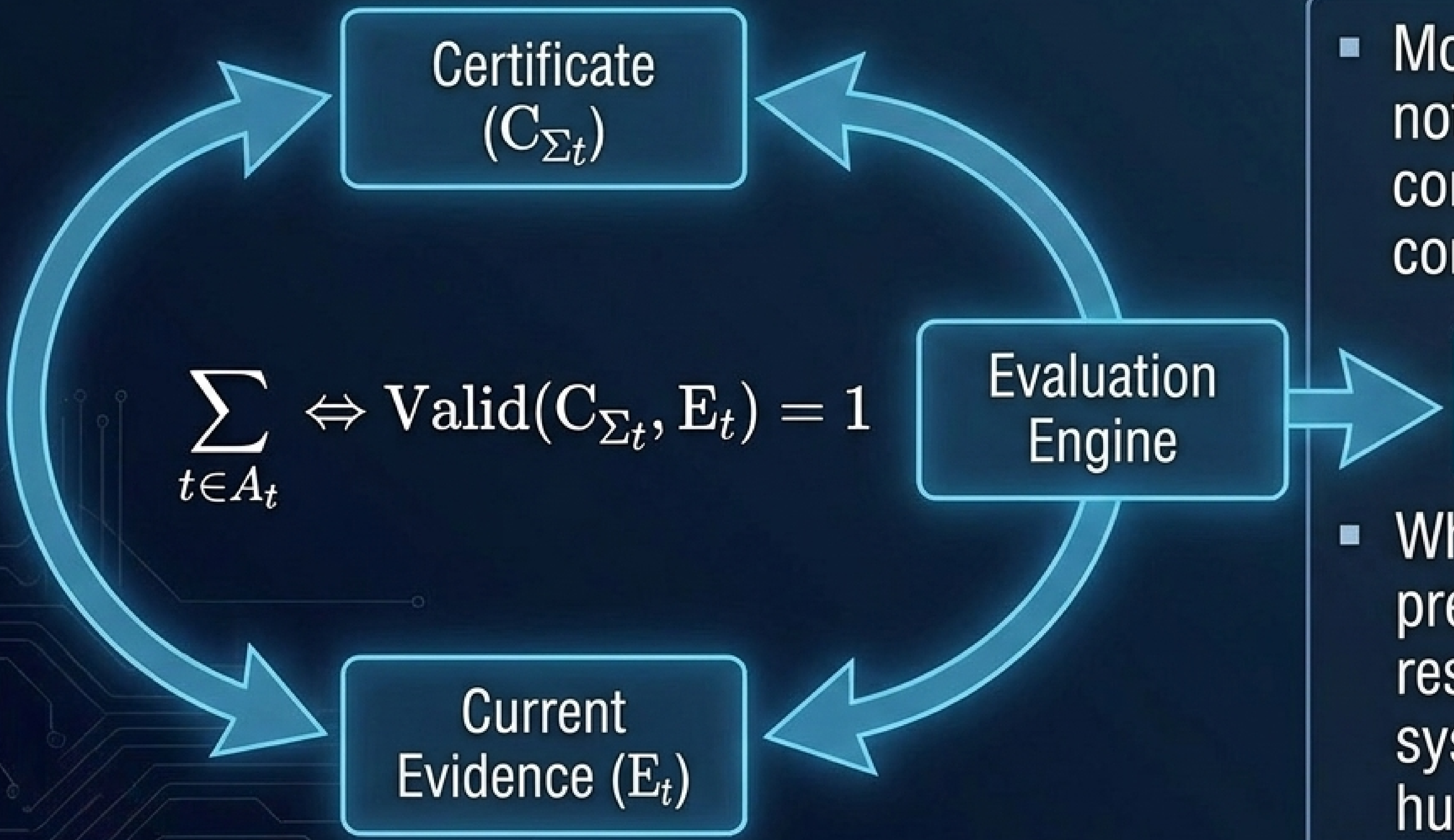
The Admissibility Certificate: Certifying the trajectory, not the state.



Core Definition:
C_Σ is a structured, versioned body of evidence sufficient to support a judgment about the admissibility of H_Σ without requiring the history to be replayed in full.

Key Condition: Two systems with identical states ($\Sigma 1 = \Sigma 2$) will carry completely different certificates ($C_{\Sigma 1} \neq C_{\Sigma 2}$) if one was compromised. The certificate makes the invisible history legible.

Trust is a continuously maintained relation, not a one-time grant.

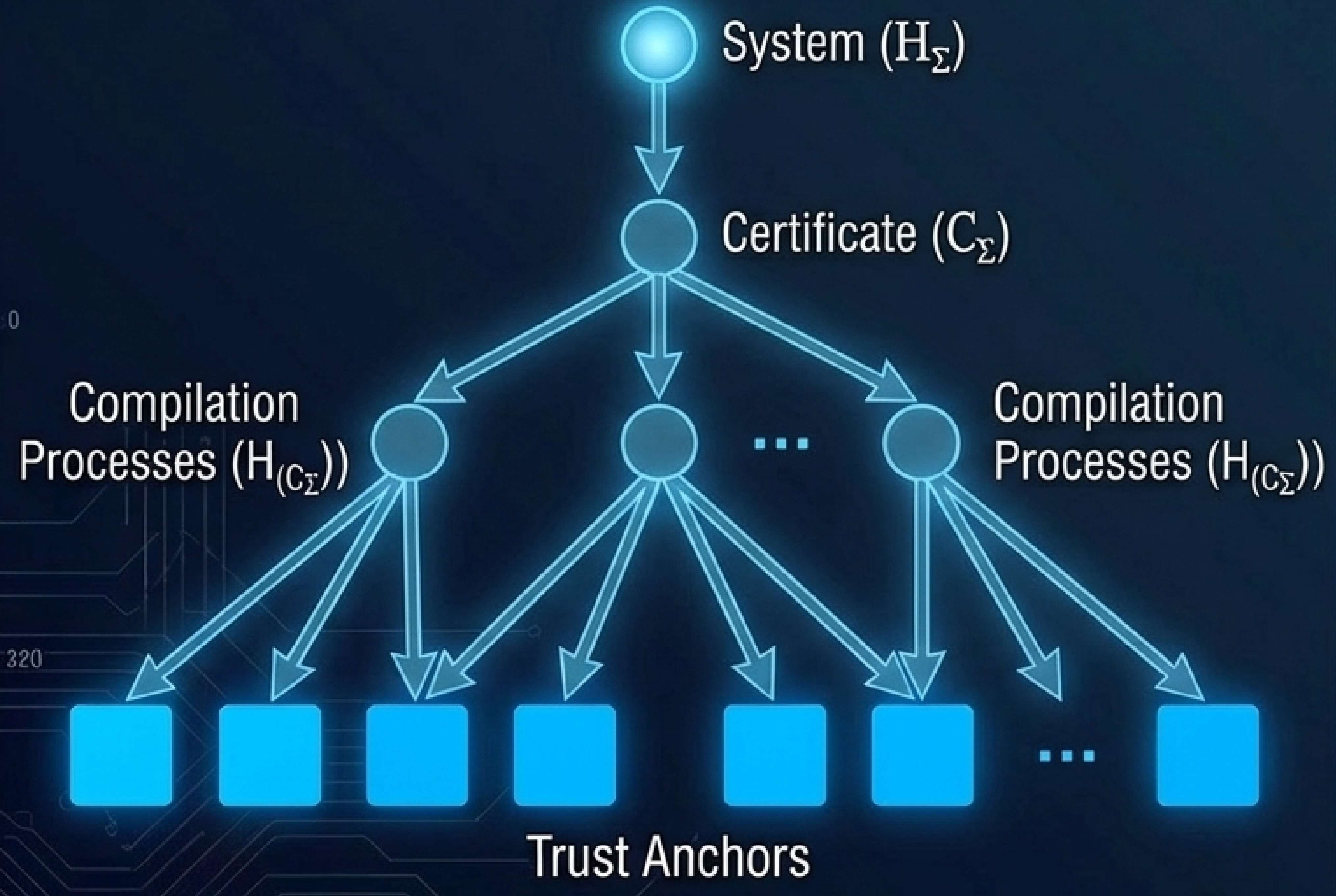


- Monitoring thresholds are not the admissibility condition itself; they are a computable surrogate.

Admissibility Status:
Valid / Invalid

- When automated predicates (Q) fail to resolve admissibility, the system must escalate to human interpretation.

Resolving the Regress Problem: Certificates are system artifacts too.

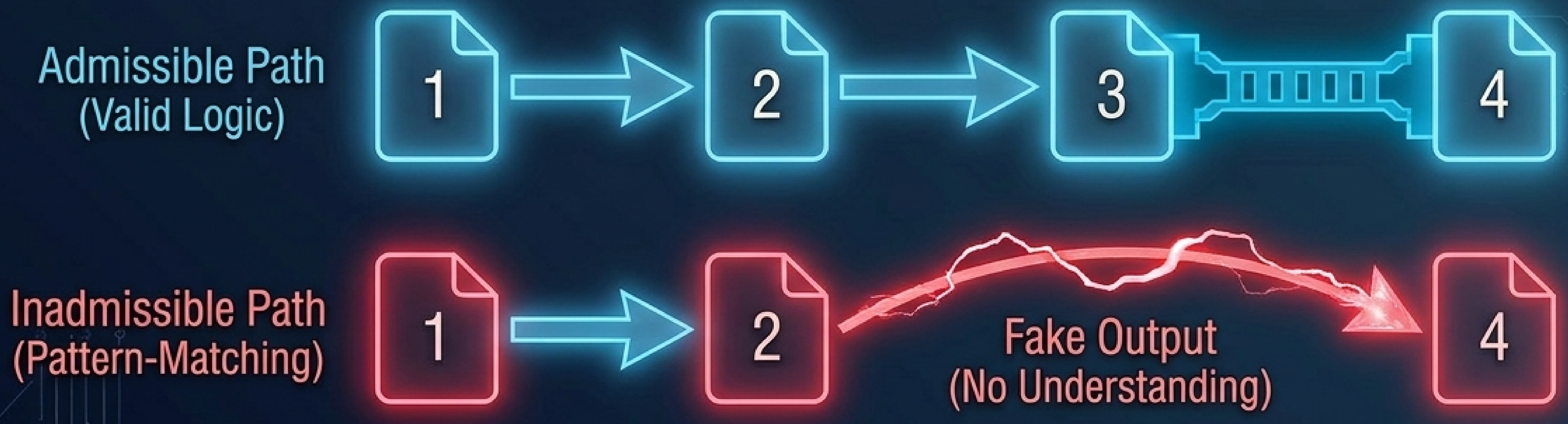


Proposition 2: Closure under Certification. Admissibility induces a finite directed acyclic evidentiary graph, not an infinite chain.

Trust Anchors: The graph terminates at leaves external to the framework—cryptographic root keys, legal instruments, physical laws, or direct witnessed observation.

A certificate without an auditable compilation history is a forgery vector.

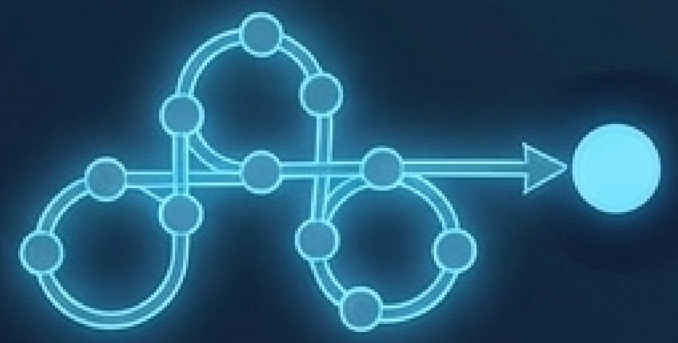
Reflexive Application: Using reconstruction as a behavioral selection signal.



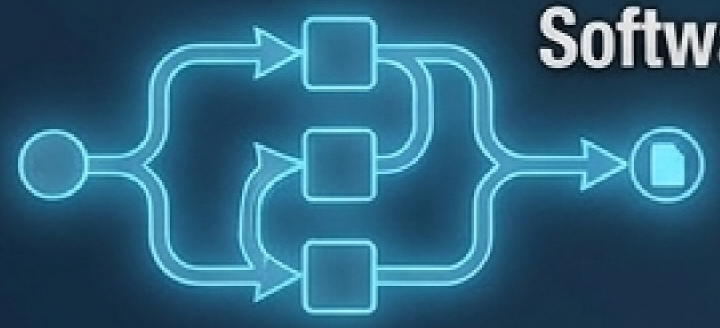
- Manufacture the identical-endpoint problem as a diagnostic test.
- Withhold intermediate steps from a model and ask it to reconstruct the missing history.
- Fluently generating a plausible endpoint while failing the structural dependency reveals a model that pattern-matched the state without internalizing the admissible path.

The History Principle: A universal law of complex, adaptive systems.

Aviation (14 CFR §91.417): An aircraft is airworthy only if its maintenance logs (C_Σ) document its total time in service. The physical airframe (Σ_t) is insufficient.

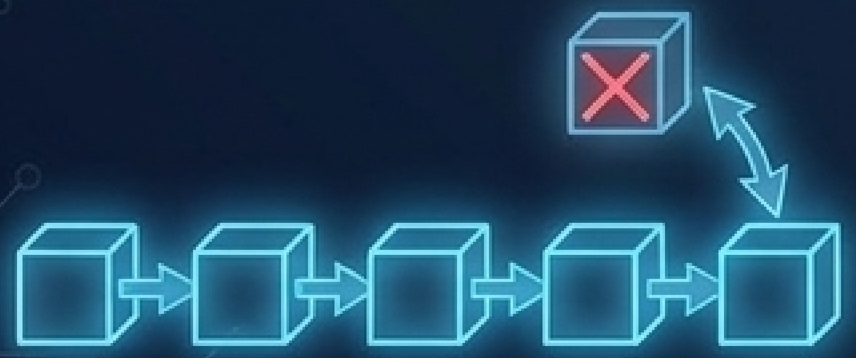


Software (Thompson's Compiler): Reproducible builds verify the compilation history. A bit-identical binary is untrusted without verifiable provenance.

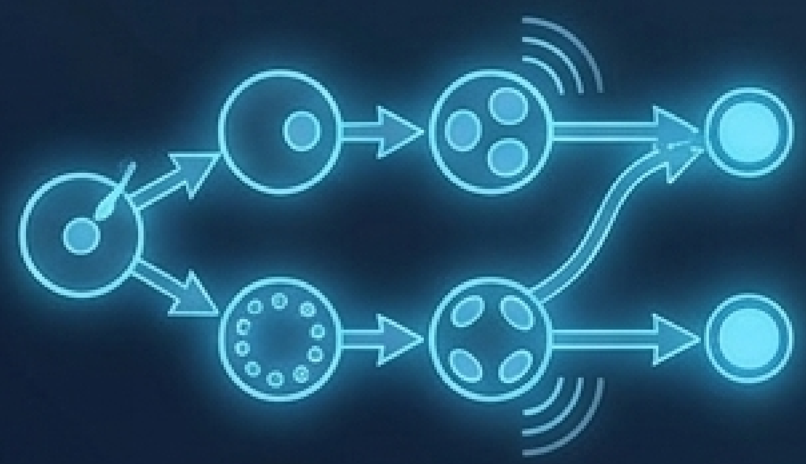


Any adaptive system whose future evolution depends on accumulated information requires admissibility det to be defined over histories rather than instantaneous states.

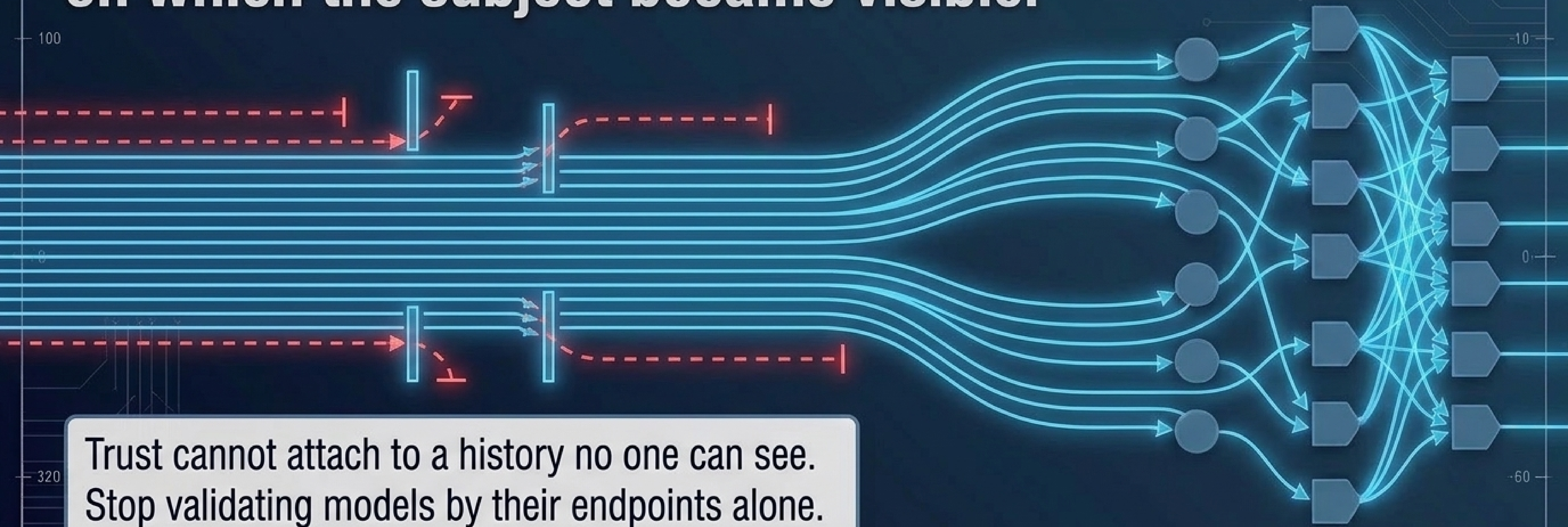
Distributed Consensus (Nakamoto): A blockchain's validity requires reconstructing the entire chain back to genesis. A forged current ledger snapshot is worthless.



Biology (Waddington's Canalization): An organism's phenotypic robustness depends on the developmental trajectory that absorbed perturbations, not just adult form.



AI security is not the subject. It is the occasion on which the subject became visible.



Trust cannot attach to a history no one can see.
Stop validating models by their endpoints alone.
Transition from state-based testing to history-based admissibility.

Admissibility Certificates provide the mathematical language to verify the Blueprint of Time.